

A185 · PhilharmonicDB: Inferring Functional Modules Across the Tree of Life

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For humans and a handful of model organisms, a sufficient quantity of experimental data (about pairs of proteins that physically bind in the cell) is available to allow powerful computational network-science inference methods to be exploited to predict protein function, protein pathways, or genes and pathways most likely to be involved in disease. However, moving outside that small set of organisms, there are typically minimal or no to no experimental interaction data available. We tap into the power of generative AI and network science to generate a set of putative functional modules for an initial 100 species of model and non-model organisms across the eukaryotic tree of life. These modules are created by our PHILHARMONIC pipeline (RECOMB 2025); the foundation of which is a lightweight sequence-based protein language model-driven parallel pipeline for fast genome-wide inference of predicted protein-protein interactions (PPIs). Although our AI inference methods sacrifice wet-lab levels of experimental accuracy for speed to enable genome-scale prediction, they create only clusters of proteins that are well supported by multiple predictions, in a way that leads to demonstrably functionally-meaningful clusters, even for organisms that are very evolutionarily distant from humans and well-studied model organisms. The functional modules of our philharmonicDB, with 100 species so far and growing, supported by the NSF AI institute for Foundations of Machine Learning (IFML) resources, is intended to be a FAIR publicly-available resource to illuminate under-studied eukaryotic genomes and democratize comparative network biology research.

PhilharmonicDB: Inferring Functional Modules Across the Tree of Life

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We previously developed PHILHARMONIC [Sledzieski et al.(2025)], an AI-based method for predicting functional networks in non-model organisms. PHILHARMONIC was originally validated using gold-standard data from the fruit fly *D. melanogaster*, and was shown to give new insights that correlated with an independent set of gene expression experiments concerning genes involved in temperature sensitivity in the coral species *P. Damicornis* and the coral algae symbiont *C. goreau*. However, the true potential of PHILHARMONIC lies in running the pipeline over many organisms, aiming for diverse coverage of the eukaryotic tree of life. The resulting resource, PhilharmonicDB, currently clusters 114 proteomes and is growing. We selected a broad range of eukaryotic genomes with annotations available from the National Center for Biotechnology Information (NCBI) Assembly and Protein databases. We specifically aimed to select species that are model organisms, important to human health or agriculture, or represent diverse branches of the eukaryotic tree, with addition consideration of genome quality statistics used to prioritize species within clades. Our database resource currently includes representatives of all major eukaryotic clades and metazoan phyla with sequencing coverage. We also include agricultural and food species (e.g., rice, cows, honeybees, shrimp, and mushrooms). Further, we prioritized several major eukaryotic pathogens of humans: *Plasmodium*/malaria, *Toxoplasma*, *Cyclospora*, *Entamoeba Naegleria*, *Trypanosoma* *Leishmania*, *Giardia*, and *Trichomonas*. We have also included some animal parasites, such as the pork roundworm, pork tapeworm, leech, blood fluke, and old world hookworm as well as animal carriers of pathogens, such as the malaria mosquito and black-legged tick.

In order to substantially scale up PHILHARMONIC across the tree of life, we apply a new parallel D-SCRIPT implementation that is designed for efficiently evoking D-SCRIPT over all pairs of an organism's complete proteome [Schäffer et al.(2025)] and also need to take advantage of large GPU resources that we were able to leverage thanks to the UT-Austin Center for Generative AI and the NSF AI Institute for Foundations of Machine Learning (IFML). We perform some preliminary analysis where our ultimate goal is to show how the AI-assisted networks we are generating in the PhilharmonicDB database can currently be used to infer "rules of life" style insights into the structure and organization of Eukaryotic PPI networks across species.

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